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United States Patent Application Publication

20250281207

Kind Code

A1

Publication Date

September 11, 2025

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Dual Head Elbow Plate

Abstract

A bone plate for stabilizing a bone includes a unitary plate having a first portion, a second portion, and a third portion with at least one compression hole on each portion. Each of the second and third portions are attached to the first portion and branch away from each other. The compression holes allow compression along three different directions. Also provided is a bone plating system including a single plate having a first portion, a second portion, and a third portion with at least one locking compression hole on each portion; a non-locking bone screw; and a locking bone screw. Additionally provided is a method of stabilizing a bone.

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Family ID: 96948271

Appl. No.: 18/652290

Filed: May 01, 2024

Foreign Application Priority Data

GR 20240100174

Mar. 08, 2024

Publication Classification

Int. Cl.: A61B17/80 (20060101)

U.S. Cl.:

CPC A61B17/8014 (20130101); A61B17/8052 (20130101); A61B17/8061 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of the filing date of Greek patent application Ser. No. 20240100174, filed Mar. 8, 2024, the disclosure of which is hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] Bone plates are commonly used for osteosynthesis applications, in particular for the treatment of bone fractures and bone stabilization. A typical fracture fixation procedure involves fixing a single bone plate to bone portions on either side of a fracture. A reduction of the fracture may also be performed, either utilizing secondary tools or through the use of compression holes in the bone plate. The latter may include the fixing of the bone plate on one side of the fracture and inserting a screw into a compression hole located on the opposite side. A typical compression hole is designed to cause a translation of the screw therein, which in turn translates the underlying bone portion to thereby reduce the fracture.

[0003] Certain fractures and/or anatomical features pose certain issues to this type of procedure. For instance, elbow fractures are often challenging for orthopedic surgeons to treat due to difficulty in achieving bone stabilization. A common surgical approach for bone plating to treat an elbow fracture includes a mid-line posterior incision that exposes the distal humerus medially, posteriorly, and laterally. Such an incision typically allows for the placement of two bone plates to stabilize the bone.

[0004] While using dual bone plates to treat elbow fractures is common, this dual plating still poses many challenges such as limited linear compression options, increased operation time, and excessive drilled holes and screws on the bone. Other drawbacks during operation include the crowding of many different plates in the surgical tray to ensure the plate required to fix the fracture is provided. Additionally, the overlap of independent plate shafts can limit some of the dual plating combinations e.g., a combination of a posterolateral plate and a posteromedial plate is not possible.

[0005] Unitary elbow plates used to treat elbow fractures are limited to specific configurations (e.g., pure posterior shafts and articular extensions), and such plates lack the ability to allow multi-directional compressions along different axes. Many such plates also often lack the ability for variable angle screw placement. Consequently, these plates may not be an ideal option for bone stabilization.

[0006] Thus, there is a need for a dual headed plate having advanced compression options with a faster and easier application than two independent plates that eliminates known drawbacks.

BRIEF SUMMARY OF THE INVENTION

[0007] The present invention addresses the aforementioned drawbacks by providing a unitary dual headed plate with multi-directional compression along different axes. The unitary plate includes a Y-shaped plate with at least one compression hole on each of the three ends. These compression holes are configured to allow advanced compression options during reduction of the fracture, especially in the context of an elbow plate. The plates according to the present invention provide the surgeon with options heretofore not seen in elbow plates, including the ability to treat complex fractures.

[0008] In accordance with an aspect of the present disclosure, a bone plate for stabilizing a bone fracture includes: a unitary plate having a first portion, a second portion, and a third portion, each of the second and third portions being attached to the first portion and branching away from each other; at least one compression hole on each portion of the unitary plate; and compression holes that are configured to allow compression along three different directions.

[0009] Embodiments of this first aspect may include a Y-shaped unitary plate.

[0010] In a further aspect, at least one compression hole is a locking compression hole.

[0011] In yet a further aspect, the bone plate includes a non-locking hole.

[0012] In yet a further aspect, the bone plate includes a locking hole.

[0013] In yet a further aspect, the compression hole is configured to allow independent compression along the first portion, the second portion, and the third portion.

[0014] In yet a further aspect, the locking compression hole is configured to accept the locking screw and the non-locking screw at varying angles with respect to a central axis of the hole.

[0015] In yet a further aspect, the second portion and the third portion branch away from each other at an angle of 90 or 180 degrees.

[0016] In accordance with another aspect of the present disclosure, a bone plating system for stabilizing a bone fracture includes: a single plate having a first portion, a second portion, and a third portion; at least one locking compression hole on each portion of the single plate; a non-locking bone screw capable of being received through one of the locking compression holes and into a bone; and a locking bone screw capable of being received through one of the locking compression holes and into the bone. The first portion splits into the second and third portions and the locking compression holes include longitudinal axes along a bone surface.

[0017] Embodiments of this second aspect may include a single plate configured to allow multi-directional compression along the longitudinal axes on the bone surface.

[0018] In a further aspect, the single plate is configured to allow multi-directional compression along the longitudinal axes on the bone surface at an angle of 50 to 160 degrees.

[0019] In another aspect, the single plate is configured to allow multi-directional compression along the longitudinal axes on the bone surface towards a single point on the bone.

[0020] In yet another aspect, the single plate is configured to allow multi-directional selective compression and decompression along the longitudinal axes on the bone surface.

[0021] In yet another aspect, the single plate is configured to allow varying compression length along the longitudinal axes on the bone surface.

[0022] In yet another aspect, the single plate is configured to allow lagging of bone fragments in a crisscross function.

[0023] In accordance with another aspect of the present disclosure, a method of stabilizing a bone includes: placing a plate on a bone; inserting a first non-compression bone screw through a first hole in a second portion of the bone plate; inserting a second non-compression bone screw through a first hole in a third portion of the plate; inserting a first compression bone screw through a first hole in a first portion of the plate and compressing a first portion of the bone by threading the first compression bone screw into the bone; inserting a second compression bone screw through a second hole in the second portion of the plate, removing the first non-compression bone screw, and compressing the second portion of the bone by threading the second compression bone screw into the bone; and inserting a third compression bone screw through a second hole in the third portion of the plate, removing the second non-compression bone screw, and compressing the third portion of the bone by threading the third compression bone screw into the bone, wherein the non-compression bone screw includes a non-locking screw and a locking screw.

[0024] Embodiments of this third aspect may include threading the compression bone screw into the bone at an angle through the locking compression hole.

[0025] In a further aspect, the compression bone screw includes the non-locking bone screw.

[0026] In yet a further aspect, the plate is manufactured via an additive manufacturing process.

[0027] In yet a further aspect, the bone is a distal humerus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] A more complete appreciation of the subject matter of the present invention and of the various advantages thereof can be realized by reference to the following detailed description in which reference is made to the accompanying drawings in which:

[0029] FIG. 1A is a top view of a bone plate in accordance with one embodiment of the present invention.

[0030] FIG. 1B is a perspective view of the bone plate of FIG. 1A placed on a distal portion of a humerus bone.

[0031] FIG. 2 is a bottom view of the bone plate of FIG. 1A.

[0032] FIG. 3 is another perspective view of the bone plate of FIG. 1A.

[0033] FIG. 4A is a left side elevation view of the bone plate of FIG. 1A.

[0034] FIG. 4B is a right side elevation view of the bone plate of FIG. 1A.

[0035] FIG. 5A is a top view of the bone plate according to another embodiment of the present invention.

[0036] FIG. 5B is a perspective view of the bone plate of FIG. 5A placed on a distal portion of a humerus bone.

[0037] FIG. 6 is a top view of the bone plate according to another embodiment of the present invention showing multi-directional compression towards a single point.

[0038] FIG. 7 is a top view of the bone plate of FIG. 6 showing varying compression length.

[0039] FIG. 8A is a top view of the bone plate according to another embodiment of the present invention.

[0040] FIG. 8B is a perspective view of the bone plate of FIG. 8A placed on a distal portion of a humerus bone.

[0041] FIG. 9A is a top view of the bone plate according to another embodiment of the present invention.

[0042] FIG. 9B is a perspective view of the bone plate of FIG. 9A placed on a distal portion of a humerus bone.

[0043] FIG. 10A is a top view of the bone plate according to another embodiment of the present invention.

[0044] FIG. 10B is a perspective view of the bone plate of FIG. 10A placed on a distal portion of a humerus bone.

[0045] FIG. 11A is a top view of the bone plate according to another embodiment of the present invention.

[0046] FIG. 11B is a perspective view of the bone plate of FIG. 11A placed on a distal portion of a humerus bone.

[0047] FIG. 12A is a top view of the bone plate of FIG. 11A showing the crisscross placement of screws to achieve a lagging function.

[0048] FIG. 12B is a perspective view of the bone plate of FIG. 12A placed on a distal portion of a humerus bone.

DETAILED DESCRIPTION

[0049] In describing the preferred embodiments of the invention, specific terminology will be used for the sake of clarity. However, the invention is not intended to be limited to any specific terms used herein, and it is to be understood that each specific term includes all technical equivalents, which operate in a similar manner to accomplish a similar purpose. In the drawings and in the description which follows, the term “proximal” means closer to the heart and the term “distal” means more distant from the heart. The term “anterior” means towards the front part of the body or the face and the term “posterior” means towards the back of the body. The term “medial” means toward the midline of the body and the term “lateral” means away from the midline of the body. The term “posterolateral” means towards the back of the body and away from the midline of the body. The term “posteromedial” means towards the back of the body and towards the midline of the body.

[0050] Certain bone plates of different bone plating systems are disclosed herein. These plates are designed to be anatomically specific, thereby ensuring accurate placement on typical bones. Such plates have also been designed in various sizes to be useable in connection with as many differently

sized patients as possible. The particular designs shown have been designed using Stryker Orthopedics Modeling and Analytics (“SOMAR”) software. This resulted in evidence-based screw hole placement designed to offer a wide range of trajectory options for particular anatomy, as well as anatomical fit, creating contoured plates with left and right specific options. It is contemplated to further refine bone plating system to include plates for a particular type of patient, for instance, male/female and ethnic-specific. It is also contemplated to provide patient-specific plates in accordance with the present invention.

[0051] While certain of the plates disclosed herein are shown in the accompanying drawings as being anatomically correct for placement on a specific bone (e.g. a left elbow), it is to be understood that the plates of the present invention could be designed for placement on a different bone (e.g., a right elbow). For instance, the left and right versions of a plate according to the present disclosure may simply be mirror images of each other. Likewise, the plates disclosed in the present application can be tailored to fit on other bones entirely—e.g., the radius, the ulna or another bone such as the tibia.

[0052] Referring to FIG. 1A, a bone plate according to an embodiment of the present disclosure includes a unitary plate **100** having a first portion **102**, a second portion **104**, and a third portion **106**. First portion **102** is positioned over the posterior side of humerus as shown in FIG. 1B. Second portion **104** and third portion **106** are attached to first portion **102** and branch away from each other to define a Y-shaped plate **100**. Each portion of plate **100** includes at least one compression hole. The inclusion of compression holes on each portion allows for compression along three different directions along a bone surface. This is particularly important for injuries like elbow fractures that include more than one fracture of the bone, which will be discussed in more detail below.

[0053] The plates of the present invention can include compression holes in any form as are known in the art. For instance, as shown in FIG. 1A, plate **100** may include compression holes in the form of a locking compression hole **108** or a non-locking compression hole **112**. Both types of compression holes may be used to apply compression along bone surface to reduce a fracture, with locking compression holes **108** also providing a locking function if desired. In this way, these holes may accept both locking and non-locking screws. The inclusion of compression holes on each portion allows independent compression on bone surface along first portion **102**, second portion **104**, and third portion **106** of unitary plate **100**. Indeed, plate **100** includes multiple compression holes on each portion that are oriented opposite to each other. Depending upon what type of injury is being treated, a surgeon can select one or more compression holes to properly reduce fracture(s) of the bone.

[0054] Bone plate **100** may also include one or more fixation holes, such as non-locking holes **110**. In other embodiments, these fixation holes could be locking holes or any other type of hole. Moreover, plates in accordance with the present invention may include any combination of the compression and fixation holes. In one embodiment, non-locking screws can be angled (e.g., up to 15 degrees in each direction for a total range of 30 degrees) to provide the flexibility to position screws based on patient needs. Locking holes may receive bone screws with threaded heads that engage threads in locking hole as to lock screw into plate. These fixation holes could also be designed such that both locking and non-locking screws can be received in a single hole. For instance, fixation holes in accordance with the present invention may be like those disclosed in U.S. Pat. Nos. 6,322,562 and 11,039,865, the disclosures of which are hereby incorporated by reference herein. Of course, other designs are also contemplated outside of just those patents. Locking compression holes **108** may include a similar fixation hole portion as do non-locking holes **110**.

[0055] The dual headed plate of the present disclosure may be applied as treatment of distal humeral fractures as well as other joints. For instance, it is also contemplated to utilize similar plate designs in connection with distal femur or proximal tibia fractures. Plates in accordance with the

present invention may be manufactured via an additive manufacturing process and specific designs can be based upon a specific patient. For instance, preoperative scans of a bone to be treated with a plate of the present invention can be utilized to generate patient-specific designs directed to the specific procedure to be performed. This can include not only plate shape, but also screw hole placement for optimal fracture fixation. Moreover, each portion of plates in accordance with the present invention can be designed to sit on different portions of the bone to be repaired. For instance, FIGS. **1B**, **5B**, **8B**, **9B**, **10B**, and **11B** show unitary plates placed on a distal portion of a humerus bone. In this regard, it is contemplated for plates in accordance with the present invention to be designed for placement in specific orientations (e.g., anterior, medial or lateral). For instance, FIGS. **1A**, **5A**, and **6** show unitary plates intended for placement in posterior, posterolateral, and posteromedial positions.

[0056] Reference will now be made to various embodiments of bone plates of the present disclosure. Wherever possible, the same or like reference numbers will be used throughout the drawings to refer to the same or like features within a different 100-series of numbers. The differences between different embodiments will be focused on below, with the understanding that other portions of the plate may be similar or even identical.

[0057] Beginning with FIG. **1A**, plate **100** includes a combination of posterior **102**, posterolateral **104**, and posteromedial **106** portions. Whereas, independent posterolateral and posteromedial plates are unable to be combined due to shaft overlap, plate **100** makes this combination possible. As shown in FIG. **1A**, plate **100** includes at least two locking compression holes **108** on each portion of plate. Plate **100** also includes at least one of or a combination of non-locking holes **110**, locking holes, or elongate **112** holes. Plate **100** allows selective multi-directional compression along longitudinal axes **114** and/or decompression along longitudinal axes **118** on bone surface as needed to reduce fractures and restore normal patient anatomy. Longitudinal axes **114** show different directions along each portion of plate **100** which allow for reduction of multiple bone fragments. This is due to the inclusion of differently oriented compression holes **108**. Thus, when plate **100** is initially held into place via initial non-locking or locking screws, multi-directional compression and/or decompression can be applied via screws inserted in different locking compression holes **108**. Further locking and non-locking screws can then be inserted into plate **100** to hold bone fragments in place after reduction is complete. FIG. **1B** illustrates plate **100** in place on a distal portion of the humerus, while FIGS. **2-4B** show different views of the plate. FIG. **5A** shows plate **200**, which is similar to that of FIG. **1A**. In particular, plate **200** includes both non-locking holes **210** and elongate holes **212**. Elongate holes **212** may also be used to apply compression along bone surface.

[0058] A further plate **300** is shown in FIG. **6**, the design of which is also similar to that of FIG. **1A**. However, plate **300** includes a single locking compression hole **308** on each of portions **304**, **306**. First portion **302**, on the other hand, includes two locking compression holes **308**. As each locking compression hole **308** faces in the same direction, they allow for multi-directional compression along longitudinal axes **314** on bone surface when screws are inserted through holes and into bone to reduce bone fragments. Plate **300** also allows multi-directional compression along longitudinal axes **314** on bone surface at projected angles **316** ranging from 50 to 160 degrees on a projection plane along posterior view of bone. As shown in FIG. **6**, plate **300** may also allow multi-directional compression along longitudinal axes **314** on bone surface toward a single point, region of clinical importance, or compression towards a metaphyseal area of bone to reduce bone fragments. Multi-directional compression applied on bone surface may be equal along each longitudinal axis **314**. As shown in FIG. **7**, plate **300** also may allow varying multi-directional compression along longitudinal axes **314** on bone surface as appropriate for proper reduction of bone fragments. Multi-directional compression applied on bone surface may be different along each longitudinal axis **314**.

[0059] It is possible to vary the design of first, second and third portions of plates in accordance

with the present disclosure to allow for reduction of complex bone fractures that previously could have only been accomplished using dual plating systems. As shown in FIG. 8A, plate **400** includes first portion **402**, second portion **404**, and third portion **406**. Second portion **404** and third portion **406** are attached to first portion **402** and branch away from each other in a Y-shape in a similar fashion as in the above-discussed embodiments. First portion **402** is positioned over the posterior side of humerus. Second portion **404** and third portion **406** of plate **400** branch away from each other at a 90 degree angle. Second portion **404** and third portion **406** are positioned in a perpendicular configuration on bone. As such, plate **400** includes coverage of posteromedial and lateral portions of bone when reducing bone fragments. FIG. 8B illustrates plate **400** on a distal portion of a humerus bone.

[0060] As shown in FIG. 9A, the design of plate **500** is similar to that of FIG. 8A. First portion **502** is positioned over posterior side of humerus. Second portion **504** and third portion **506** of plate **500** branch away from each other at a 90 degree angle. Second portion **504** and third portion **506** are positioned in a perpendicular configuration on bone. As such, plate **500** includes coverage of posterolateral and medial portions of bone when reducing bone fragments. FIG. 9B illustrates plate **500** on a distal portion of a humerus bone.

[0061] As shown in FIG. 10A, the design of plate **600** is also similar to that of FIG. 8A. First portion **602** is positioned over medial side of humerus. Second portion **604** and third portion **606** of plate **600** branch away from each other at a 180 degree angle. Second portion **604** and third portion **606** are positioned in a parallel configuration on bone. As such, plate **600** includes coverage of medial and lateral portions of bone when reducing bone fragments. FIG. 10B illustrates plate **600** on a distal portion of a humerus bone.

[0062] As shown in FIG. 11A, the design of plate **700** is also similar to that of FIG. 8A. First portion **702** is positioned over the posterior side of humerus. Second portion **704** and third portion **706** of plate **700** branch away from each other at a 180 degree angle. Second portion **704** and third portion **706** are positioned in a parallel configuration on bone. As such, plate **700** includes coverage of posterior, lateral, and medial portions of the bone when reducing bone fragments. FIG. 11B illustrates plate **700** on a distal portion of a humerus bone.

[0063] As shown in FIG. 12A, plate **700** includes holes that allow for lagging of bone fragments in a crisscross function rather than a traditional single lagging originating from a linear plate. Crisscross lagging of bone fragments allows multiple bone fragments to be compressed together by inserting screws through each fragment at an angle. For instance, medial and lateral fragments are compressed towards lateral and medial distal portions of the plate respectively. In one embodiment, non-locking screws may be inserted into non-locking holes **710** at an angle in a range of 30 degrees on second portion **704** and third portion **706** of plate **704**. FIG. 12B illustrates plate **700** on distal portion of humerus bone showing crisscross lagging of bone fragments. Crisscross lagging of bone fragments requires direct lateral and direct medial distal plate portions as shown in bone plate **700** and bone plate **600**. Bone plate **600** of FIG. 10A also includes non-locking holes **610** on second portion **604** and third portion **606** that allow for lagging of bone fragments in a crisscross function.

[0064] In accordance with another aspect of the present disclosure, a bone plating system includes a single plate including a first portion, a second portion, and a third portion. First portion splits into second and third portion forming Y-shaped single plate. Each portion of single plate includes at least one locking compression hole. Bone plating system also includes a non-locking bone screw and a locking bone screw capable of being received through locking compression holes and threaded into bone.

[0065] In accordance with yet another aspect of the present disclosure, a method of stabilizing a bone includes placing a bone plate on bone, inserting locking or non-locking screws into bone plate, inserting compression screws into bone plate, and threading compression screws into bone. Bone compression occurs by tightening compression screws in first, second, and third portion of plate. During compression of each plate portion the other two portions are firmly mounted to bone

by at least one locking or non-locking screw. Any locking or non-locking screws that were previously added in portion of plate are removed before tightening compression screw in same portion to allow bone compression. The sequence of inserting and threading compression screws depends on which portion of bone needs to be compressed first to address bone fracture. Bone screws may be threaded into bone at an angle through locking compression holes, locking holes, non-locking holes, and elongate holes.

[0066] In a specific embodiment, the method may include inserting locking or non-locking screws into second and third portions of plate and then inserting a compression screw into a first portion of plate. This results in the compression of at least one fracture of bone. Thereafter, a compression screw can be inserted into second portion after removing the previously inserted locking or non-locking screw. This results in the compression of another fracture. Finally, a compression screw can be inserted into third portion after removing the previously inserted locking or non-locking screw to compress yet another fracture. Compression screws include non-locking screws.

[0067] In further detail, the method of stabilizing bone typically includes a mid-line posterior incision that exposes distal humerus medially, posteriorly, and laterally. This type of incision allows for insertion of bone plate. The method of stabilizing bone first includes taking scans of bone and evaluating which plate structure will be most efficient to stabilize bone and reduce bone fragments. The type of bone fracture, individual bone anatomy, and surgeon preference are considered when choosing bone plate structure. Plates would primarily be applied in complete articular elbow fractures (type 13C fractures as per AO/OTA classification). This includes simple articular, simple metaphyseal (type 13C1 as per AO/OTA classification), simple articular, fragmentary metaphyseal (type 13C2 as per AO/OTA classification) and multifragmentary articular metaphyseal (type 13C3 as per AO/OTA classification). In clinical practice, a specific fracture pattern may indicate a specific dual plate selection.

[0068] After bone anatomy and bone fracture are evaluated and an appropriate bone plate is determined, a suitable bone plate is positioned over bone fracture, for instance plate **100**. The bone plate is fixed to one spot on bone surface using bone screws. Elongated holes **112** and non-locking holes **110** may receive bone screws in a non-fixed state allowing for final fixation and positioning of bone plate. Exact positioning of bone plate can be adjusted to achieve desired positioning and compression along bone surface for bone fragment reduction. Elongated holes **112** also allow for adjusting positioning and orientation of bone plate because bone plate may be moved along a distance of elongated opening before final screws are fixed into place.

[0069] After bone plate is fixed into place, multi-directional compression or decompression is applied along bone surface to reduce bone fragments by inserting screws into locking compression holes **108**. Once reduction is complete, final locking and non-locking screws may be inserted into locking screw holes, non-locking holes **110**, and elongated holes **112** to fix bone fragments into place. Locking screws can be angled within non-locking holes **110** and locking compression holes **108** (e.g., up to 15 degrees in each direction for a total range of 30 degrees) depending upon patient anatomy.

[0070] Bone plate may be manufactured via an additive manufacturing process. Additionally, bone plates and bone screws may be made of a biocompatible material such as a polymer e.g., reinforced polymer composites; a titanium alloy e.g., Ti6Al4V; stainless steel e.g., **316L**; or other metals and metal alloys, such as Ti and CoCr.

[0071] Unitary plate systems allow for a faster and easier application of bone plating to stabilize bone. Further, less drilled holes and screws are needed on bone along plate shaft in particular with the combination of non-locking holes and locking holes along bone plate that allow for exact positioning of bone plate before fixing it into place. Plates according to the present disclosure also allow for dual plating combinations that are not possible with dual plates such as a combination of posterolateral plates and posteromedial plates.

[0072] Although the invention herein has been described with reference to particular embodiments,

it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. For example, features described in relation to one particular embodiment may be combined with features of other embodiments described herein. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined in the appended claims.

Claims

1. A bone plate comprising: a unitary plate including a first portion, a second portion, and a third portion, each of the second and third portions being attached to the first portion and branching away from each other; and at least one compression hole on each portion of the unitary plate, wherein the compression holes are configured to allow compression along three different directions.
2. The bone plate of claim 1, wherein the unitary plate is Y-shaped.
3. The bone plate of claim 1, wherein at least one compression hole is a locking compression hole.
4. The bone plate of claim 1, further comprising a non-locking hole.
5. The bone plate of claim 1, further comprising a locking hole.
6. The bone plate of claim 1, wherein the compression hole is configured to allow independent compression along the first portion, the second portion, and the third portion.
7. The bone plate of claim 3, wherein the locking compression hole is configured to accept the locking screw and the non-locking screw at varying angles with respect to a central axis of the hole.
8. The bone plate of claim 1, wherein the second portion and the third portion branch away from each other at an angle of 90 or 180 degrees.
9. A bone plating system comprising: a single plate including a first portion, a second portion, and a third portion; at least one locking compression hole on each portion of the single plate; a non-locking bone screw capable of being received through one of the locking compression holes and into a bone; and a locking bone screw capable of being received through one of the locking compression holes and into the bone, wherein the first portion splits into the second and third portions, and wherein the locking compression holes include longitudinal axes along a bone surface.
10. The bone plating system of claim 9, wherein the single plate is configured to allow multi-directional compression along the longitudinal axes on the bone surface.
11. The bone plating system of claim 10, wherein the single plate is configured to allow multi-directional compression along the longitudinal axes on the bone surface at an angle of 50 to 160 degrees.
12. The bone plating system of claim 11, wherein the single plate is configured to allow multi-directional compression along the longitudinal axes on the bone surface towards a single point on the bone.
13. The bone plating system of claim 11, wherein the single plate is configured to allow multi-directional selective compression and decompression along the longitudinal axes on the bone surface.
14. The bone plating system of claim 11, wherein the single plate is configured to allow varying compression length along the longitudinal axes on the bone surface.
15. The bone plating system of claim 9, wherein the single plate is configured to allow lagging of bone fragments in a crisscross function.
16. A method of stabilizing a bone comprising the steps of: placing a plate on a bone; inserting a first non-compression bone screw through a first hole in a second portion of the bone plate; inserting a second non-compression bone screw through a first hole in a third portion of the plate; inserting a first compression bone screw through a first hole in a first portion of the plate and

compressing a first portion of the bone by threading the first compression bone screw into the bone; inserting a second compression bone screw through a second hole in the second portion of the plate, removing the first non-compression bone screw, and compressing the second portion of the bone by threading the second compression bone screw into the bone; and inserting a third compression bone screw through a second hole in the third portion of the plate, removing the second non-compression bone screw, and compressing the third portion of the bone by threading the third compression bone screw into the bone, wherein the non-compression bone screw includes a non-locking screw and a locking screw.

17. The method of claim 16, wherein the compression bone screw may be threaded into the bone at an angle through the locking compression hole.

18. The method of claim 16, wherein the compression bone screw includes the non-locking bone screw.

19. The method of claim 16, wherein the plate is manufactured via an additive manufacturing process.

20. The method of claim 16, wherein the bone is a distal humerus.
